Sharp State Complexity of Cyclic One-Pass Readout
with One Adaptive Protected Source Record

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Abstract

We study deterministic one-pass readout of a frozen LIFO stack of n distinct named records, whose independent payloads belong to a fixed alphabet of $m \geq 2$ symbols. Before reading begins, a query specifies one of the n cyclic permutations of the owners. The machine may pop the immutable original records once, keep a single actual popped (owner, payload) record in an externally accessible protected stage, and use finite control with q states. Crucially, it may choose which genuine source owner to stage as a function of previously observed payload symbols, and it can inspect the staged owner's name without rewriting the record. Every scan, query, output, and decoding phase counts towards q; there is no auxiliary scratch stack or uncounted timing channel. We prove the asymptotically optimal state complexity

$$q_{\mathrm{cyc}}(n, 1; m) = \Theta_m(m^n/n).$$

We also prove a linear-in-n state saving over the fixed-owner-stage variant. A matching constructive upper bound encodes a short input block in the identity of one *later occurring genuine* owner, and emits a small literal prefix before decoding that block; the prefix eliminates a logarithmic state overhead. The result makes an explicit model distinction between adaptive owner-identity storage, fixed-identity buffering, and classical single-use register or string-register transducers. Exhaustive small instances corroborate the construction. The theorem is model-relative and does not claim an unrestricted DPDA bound or certified historical priority over the automata literature.

Keywords. one-pass transducer; LIFO readout; adaptive buffering; finite-state complexity; cyclic permutation; named records.

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1 Scope and prior work

One-pass reordering is traditionally measured by buffer occupancy, scheduling cost, or transducer expressiveness. The present statistic is different: the number of *finite-control states* when a single protected stage retains an actual named input record. The owner's immutable name can itself transport a bounded amount of information, even when its payload is just one symbol. We ask whether that additional information capacity changes the exponential state cost of all cyclicly ordered readouts from the same frozen stack.

Praveen [1] studies streaming string/register transducers and their origin semantics.

Bojańczyk and Stefański [2] study single-use register transducers, where reading a register empties it. In contrast, our protected source record remains physically readable until its required output; its owner name can be inspected repeatedly. These models are not identified without a formal translation.

Reordering-buffer management, studied for example by Adamaszek *et al.* [3], optimizes competitive scheduling cost rather than the finite-control state count considered here. These works supply essential context, not a confirmed historical priority result for the present special cost theorem.

2 Contributions and novelty

The paper-specific contribution is an exact asymptotic state-complexity characterization for a source-typed, one-pass cyclic readout model with a single repeatedly inspectable *actual* protected source record. The lower bound accounts for the information retained in a physical owner–payload pair, while the constructive upper bound uses the identity of a *future genuine owner* to encode a short block of previously read symbols. A direct literal prefix removes the logarithmic overhead in counting all decoding states. Under payload-independent fixed-owner staging the same task has a larger state cost, yielding a linear-in-n separation between adaptive and fixed owner identity at unchanged stage capacity. Streaming/register transducer frameworks and reordering-buffer problems provide context, but their results and their models are not claimed as new or formally equivalent to the protected-source model here. A definitive historical priority claim would require a broader specialist literature comparison.

3 Machine model and main theorem

Fix an integer $n \geq 2$, a finite payload alphabet Σ with $m = |\Sigma| \geq 2$, and a set of named owners $[n] = \{1, \ldots, n\}$. The physical source is a read-once stack with top-to-bottom records

$$(1, b_1), (2, b_2), \ldots, (n, b_n), \qquad b_i \in \Sigma.$$

All m^n payload assignments are possible. At the same frozen initial stack checkpoint the environment chooses a query label $j \in [n]$ *before the scan.* The prescribed readout is the payload sequence with owners

$$\sigma_j = (j, j+1, \ldots, n, 1, \ldots, j-1). \tag{1}$$

A deterministic finite controller may read and pop each original record exactly once and cannot rewrite or reinsert it. It has q internal states and one optional protected stage. This stage is either empty or retains exactly one original, immutable record (i, b_i) already popped from the source; the controller may inspect both its genuine owner label and payload while it is present. It may stage a record in a payload-dependent manner, but cannot relabel a staged owner, change its payload, or stage two records. Each output move prints at most one payload symbol (a zero-output move is allowed). Every phase of output or scan, including query-specific modes, is included in q; there are no uncounted clocks, data registers, writable auxiliary stacks, or temporary staging positions. Owner labels on the current source record and the protected stage are observable symbols of the given typed model; the controller's transition table may depend on the fixed finite n, m.

Let $q_{\mathrm{cyc}}(n, 1; m)$ be the smallest q for one shared controller that accepts every $j \in [n]$ and produces the readout (1) for every payload assignment, with the same frozen source order.

Theorem 3.1 (Tight finite-alphabet cyclic readout frontier)**.** *For every fixed $m \geq 2$ there are positive constants c_m, C_m, n_m such that, for all $n \geq n_m$,*

$$c_m \frac{m^n}{n} \leq q_{\mathrm{cyc}}(n, 1; m) \leq C_m \frac{m^n}{n}. \tag{2}$$

The protected stage stores only one immutable genuine source record. In particular $q_{\mathrm{cyc}}(n, 1; 2) = \Theta(2^n/n)$.

4 The indistinguishability lower bound

Lemma 4.1 (Bottom-owner cut)**.** *For every n, m in the model above,*

$$q_{\mathrm{cyc}}(n, 1; m) \geq \left\lceil \frac{m^{n-1}}{1 + m(n-1)} \right\rceil . \tag{3}$$

Proof. Select query $j = n$ and fix the last owner's payload b_n to one given value. Immediately after the first $n-1$ records have been popped, but before the bottom owner can be read, the required output has not started: owner n must be output first, and its payload is still unread. In particular, an earlier first output could not be valid simultaneously for all $m \geq 2$ possible bottom payloads. We may therefore fix b_n for the counting argument without allowing any early output. All m^{n-1} assignments to the first $n-1$ bits/symbols have identical unread stack suffix, fixed output history, and identical externally controlled input position. If two assignments had the same controller state and the same stage configuration at this cut, determinism would force the same continuation and output, which is impossible. The stage has at most $1 + m(n-1)$ physical configurations: empty, or one of the $n-1$ already popped owners with one of m payload values. Hence $q[1 + m(n-1)] \geq m^{n-1}$. This proves (3). The environment has fixed the query before scanning, so it is not an uncounted post-hoc channel. □

5 A genuine-owner online encoder

For a query $j = K + 1$, the K owners preceding j in the stack must be deferred. Once j and owners $j+1, \ldots, n$ are emitted during the scan, the original deferred owners must be flushed *in increasing order* $1, \ldots, K$. We next build a one-stage encoder for such a prefix over any fixed m-symbol alphabet.

For sufficiently large K, let $d \geq 1$ be maximal with

$$m^d + 2d + \lceil \log_m(d+1) \rceil \leq K, \tag{4}$$

and put

$$r = \lceil \log_m(d+1) \rceil, \quad M = K - d - 1, \quad L = r + d + m^d - 1. \tag{5}$$

Equation (4) gives

$$L \leq M, \qquad m^r \geq d + 1. \tag{6}$$

For the finitely many K having no valid d, use an ordinary finite-control buffer storing all K symbols.

Lemma 5.1 (Online future-owner selection)**.** *For every payload word $b_1 \ldots b_K \in \Sigma^K$, one can make a single left-to-right physical pass which stores one* unaltered actually occurring source record *and at most M other payload symbols in finite control at any time. From these resources, all K deferred symbols can later be output in their original increasing-owner order. No information from an unread source record is consulted by the stage-selection rule.*

Proof. Identify Σ with $\{0, 1, \ldots, m-1\}$. Read the initial r payload symbols literally as a block A, and the following d symbols as a base-m integer $v \in \{0, \ldots, m^d - 1\}$. At this time all required selector data are already read; choose the target source index

$$i = r + d + 1 + v \leq K. \tag{7}$$

Continue scanning and accumulating the block X of the v genuine intermediate owners. When original owner i physically arrives, move *that exact record* (i, b_i) into the protected stage. Now remove the d selector symbols from the controller's buffer: the actual staged label i reversibly

represents them by $v = i - r - d - 1$. Let Y denote the later $K - i$ symbols read after staging. At the end of the deferred scan the controller holds the word $A\,X\,Y$ of length

$$|A| + |X| + |Y| = r + v + (K - i) = K - d - 1 = M,$$

while the protected stage holds (i, b_i). Just before the latest possible staging event, the controller contains at most $r + d + (m^d - 1) = L \le M$ literal symbols; at all later times it contains at most M. Thus both online and post-stage memory bounds are valid.

When the blocking owner has been passed and deferred output may start, emit the r symbols of A. Then inspect the genuine stage label i and decode the d base-m digits of $v = i - r - d - 1$ in their natural order, emitting one per step. Finally emit the $|X| = v$ buffered symbols of X, then the actual stage payload b_i (releasing the stage), then the buffered symbols of Y. The output is exactly $A\,B\,X\,b_i\,Y = b_1 \dots b_K$. The lengths of the buffered suffix and the visible stage label determine when X is exhausted; no separate unbounded counter is hidden in the schedule. □

Lemma 5.2 (All phases have $O_m(m^M)$ finite-control states)**.** *For each deferred-prefix length K and the parameters* (5)*, a deterministic implementation of Lemma 5.1 exists with at most $C'_m m^M$ controller states, where C'_m depends only on m and not on K, counting the stage-selection, source-scan, all output symbols, and decode phases.*

Proof. A finite-control buffer storing an arbitrary word over Σ of length at most H takes $\sum_{h=0}^{H} m^h \le m^{H+1}/(m-1)$ distinct payload-word states. This count already distinguishes buffer lengths, so it must not be multiplied by another factor for a separate length counter. The present implementation uses a bounded number of phase types, with source-owner and protected-owner names read directly from their physical locations. Separate mode banks for the pre-stage scan ($H = L$), the post-stage scan ($H = M$), and the direct source-output segment $j, \dots, n$ ($H = M$) therefore cost $O_m(m^M)$ states. The output of the first r literal symbols reduces buffer length by one at every step, and all those states form another $O_m(m^M)$ bank.

The delicate part is the decoded base-m owner-index block. It requires exactly d one-symbol output steps while the untouched buffer has length $M - r$. Explicitly including a distinct mode for each decode position uses at most

$$d\,m^{M-r} \le m^M$$

controller states by (6). Once decoding ends, output of X, the single stage symbol, and Y uses a constant number of modes and words of length at most $M - r$, again $O_m(m^M)$ states. All decisions about the position $|X| = v$ use the visible original staged owner label i, and so require no additional family of controller states. These counts enumerate every scan, output, and decoding phase, including the deliberate r-symbol memory reduction. Their sum is at most $C'_m m^M$. □

6 Assembly for all cyclic queries

Proof of Theorem 3.1. Lemma 4.1 proves the lower bound of order $\Omega_m(m^n/n)$. For the upper bound, fix query $j = K + 1$ before scanning, where $0 \le K \le n - 1$. Use Lemmas 5.1 and 5.2 on the prefix $1, \dots, K$, output the rest of the source $j, \dots, n$ directly during the scan, and finally emit the deferred original prefix. For fixed m, maximality in (4) implies $m^d = \Theta_m(K)$ for all sufficiently large K. Indeed, $m^d \le K$, while the failure of the inequality at $d + 1$ gives

$$K < m^{d+1} + 2(d + 1) + \lceil \log_m(d + 2) \rceil.$$

Since $d = O_m(\log K)$, the additive term is $o(K)$; hence $K = O_m(m^{d+1}) = O_m(m^d)$. Thus $m^d = \Theta_m(K)$. Since $M = K - d - 1$,

$$q_K \le C'_m m^{K-d-1} = O_m\left(\frac{m^K}{K+1}\right). \tag{8}$$

For bounded K ordinary finite-control buffering has a constant cost absorbed into this asymptotic inequality. Use disjoint, query-labelled finite-control banks for the n alternatives; these banks all operate on the *same frozen source order*, and the query's initial choice is itself counted. The total finite-control cost is bounded by

$$q_{\mathrm{cyc}}(n,1;m) \le \sum_{K=0}^{n-1} q_K = O_m\left(\sum_{K=0}^{n-1} \frac{m^K}{K+1}\right) = O_m\left(\frac{m^n}{n}\right).$$

The final estimate follows by splitting the sum at $K = \lfloor n/2 \rfloor$ and using geometric decay in its upper half. Combined with the lower bound, this proves (2). □

Remark 6.1 (Scope and necessity of the short direct prefix)**.** *Without the literal prefix A, the d successive decode phases would each retain a controller buffer with M arbitrary symbols, apparently requiring dm^M states. Taking $r = \lceil \log_m(d+1) \rceil$ and emitting A first reduces the unchanged buffer to $M-r$ symbols, so all decode phases together cost at most m^M. The identity-bearing stage remains a genuine source record; this is neither a writable data register nor an extra stack.*

7 A quantitative separation from fixed-owner staging

Suppose, in the same source-typed model, that the decision of *which owner* may be staged and whether the stage is occupied is fixed by the chosen query and source position, independently of all payload values. The protected stage can still hold the genuine symbol of the predetermined owner. Write $q_{\mathrm{cyc}}^{\mathrm{fixed}}(n,1;m)$ for the minimum controller size under this restriction.

Proposition 7.1 (Linear advantage of identity-adaptive staging)**.** *For every fixed alphabet size $m \ge 2$,*

$$q_{\mathrm{cyc}}^{\mathrm{fixed}}(n,1;m) = \Theta_m(m^{n-2}) \quad \textit{whereas} \quad q_{\mathrm{cyc}}(n,1;m) = \Theta_m(m^n/n).$$

Consequently the adaptive choice of an actual protected source owner's identity saves a factor $\Theta_m(n)$ in control states, even though the stage's symbol alphabet and physical capacity are unchanged.

Proof. At the bottom-first query's indistinguishability cut, all m^{n-1} assignments of the first $n-1$ payloads must remain distinguishable. In the fixed-owner regime the stage configuration is either always empty or has a fixed owner with at most m possible payloads, giving $q \ge m^{n-2}$. Conversely, for a query with $K = j-1 \ge 1$ deferred owners, preselect original owner K for the stage and store the first $K-1$ symbols literally in finite control. The stage record is read only when it actually appears. Once owner j and the later source owners have been printed in order, emit the $K-1$ cached symbols and then the staged Kth symbol. A constant number of mode banks with variable-length controller words uses $O_m(m^{K-1})$ states for that query, including the scan and output phases. The query $j=1$ needs only constant control. Summing the query-specific banks gives $O_m(m^{n-2})$. The final ratio with Theorem 3.1 is $\Theta_m(n)$. □

8 Finite verification and scope

The mathematical lower and upper bounds are derived symbolically; finite checks are independent consistency tests, not substitutes for universal proofs or a certificate of exact finite-state minimality. The reported tests cover binary, ternary, and quaternary alphabets on short deferred prefixes, all cyclic source/query pairs for binary instances through $n = 12$ and ternary instances through $n = 8$ (a total of 163,921 pairs), and representative memory-inequality checks for fixed alphabet sizes and prefix lengths up to 4096. Independent bounded reconstructions also checked

125,096 short-prefix payload instances; these corroborate the stated encoder identities within the tested ranges. No claim about untested parameter ranges is inferred from these counts.

The theorem is relative to a model with a genuine, repeatedly inspectable physical owner label and a query available before scanning. These features are not automatically available in a general DPDA or in a single-use atom-register transducer. A fuller specialist literature comparison would be needed to settle historical priority; the statements here are mathematical claims about the specified model.

Open Problem

Exact finite-size cyclic readout state complexity. For fixed integers $m \geq 2$ and $n \geq 2$, determine the exact minimum $q_{\mathrm{cyc}}(n, 1; m)$ for the deterministic one-pass model of this paper with one adaptive, immutable, repeatedly inspectable genuine source record, all controller phases counted, and the query supplied before scanning. Theorem 3.1 determines its asymptotic order for fixed m, but not the exact value for every finite (n, m).

Statements and Declarations

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